

Experiment 6

Objective

To explore the concepts and practices of Continuous Deployment and automated software delivery using Continuous Deployment tools.

Problem Statement

You are required to implement a **fully automated deployment pipeline** where:

- Code changes pushed to GitHub
 - Trigger a Jenkins pipeline via webhook
 - The pipeline builds and deploys the application
 - The application is published using GitHub Pages
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Tasks

Task 1: Create a Sample Web Application

1. Create a project folder named cd-demo-app
 2. Create a file index.html with basic content:
 - Include a heading: *“Continuous Deployment Demo”*
 3. (Optional) Add style.css
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Task 2: Initialize and Push to GitHub

1. Initialize a Git repository
 2. Commit the project files
 3. Create a repository in GitHub
 4. Push the code to the main branch
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Task 3: Set Up Jenkins

1. Install and launch Jenkins locally
2. Install required plugins:
 - Git plugin

- Pipeline plugin
 - 3. Create a **Pipeline Project**
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Task 4: Configure Webhook Integration

1. Use ngrok to expose Jenkins:
`ngrok http 8080`
 2. Copy the generated HTTPS URL
 3. In GitHub:
 - Navigate to **Settings → Webhooks**
 - Add webhook:
 - Payload URL: <ngrok-url>/github-webhook/
 - Trigger: *Push events*
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Task 5: Create Jenkins Pipeline

1. Configure a pipeline script in Jenkins that:
 - Clones the GitHub repository
 - Performs a basic build step
 - Copies files into a deploy/ directory
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Task 6: Configure Deployment to GitHub Pages

1. Create a gh-pages branch in your repository
 2. Generate a GitHub Personal Access Token (PAT)
 3. Store credentials in Jenkins
 4. Extend the pipeline to:
 - Push contents of deploy/ folder to gh-pages branch
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Task 7: Enable GitHub Pages

1. Go to repository **Settings → Pages**
2. Select:

- Source: gh-pages branch
3. Save configuration
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Task 8: Test Continuous Deployment

1. Modify the index.html file
 2. Commit and push changes
 3. Verify:
 - Jenkins pipeline triggers automatically
 - Deployment completes successfully
 - Changes reflect on the live website
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Expected Outcome

- Automatic deployment pipeline is successfully implemented
 - Application is accessible via a public URL
 - Changes are reflected immediately after each commit
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Submission Instructions:

1. Students must upload a single PDF file containing the report of the experiment conducted.
2. The report should contain code files, Jenkins pipeline execution screenshot, GitHub repository link, and Live website URL (GitHub Pages).
3. The PDF name should be in the form "StudentName_SAPID".